

Combination Numbers
- Pascal’s Rule: $C_n^k = C_{n-1}^k + C_{n-1}^{k-1}$. - Absorption Identity: $kC_n^k = nC_{n-1}^{k-1}$. - Trinomial Revision: $C_n^m C_m^k = C_n^{m-k} C_{n-k}^m$. - Row Sum: $\sum_{k=0}^n C_n^k = 2^n$. - Even-Odd Sums: $\sum_{k \geq 0} C_n^{2k} = \sum_{k \geq 0} C_n^{2k+1} = 2^{n-1}$. - Weighted Sums: $\sum_{k=0}^n kC_n^k = n2^{n-1}$. $\sum_{k=0}^n k^2C_n^k = n(n+1)2^{n-2}$. $\sum_{k=0}^n k^3C_n^k = n^2(n+3)2^{n-3}$. $\sum_{k=0}^m (-1)^k C_n^k = (-1)^m C_{n-1}^m$. - Hockey-Stick Identity: $\sum_{i=k}^n C_i^k = C_{n+1}^{k+1}$. - Diagonal Sum: $\sum_{k=0}^m C_{n+k}^k = C_{n+m+1}^m$. - Vandermonde’s Identity: $\sum_{k=0}^r C_n^k C_m^{r-k} = C_{n+m}^r$. - Negative Binomial Sum: $\sum_{k=0}^\infty C_{n+k-1}^{n-1} x^k = (1-x)^{-n}$.
Probability Space
Sample Space Ω: the set of all possible outcomes. σ-algebra: the collection of events. $\emptyset \in \mathcal{F}$, - If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$, - If $A_i \in \mathcal{F}$, then $\bigcup_i A_i \in \mathcal{F}$. Borel field: all intervals and σ-algebra generated from them. Probability Measure $\mathbf{P}$: a function $\mathbf{P}: \mathcal{F} \rightarrow [0, 1]$. - Nonnegativity: $\mathbf{P}(A) \geq 0, \forall A \in \mathcal{F}$, - Normalization: $\mathbf{P}(\emptyset) = 0, \mathbf{P}(\Omega) = 1$, - Countable additivity: If $A_i \in \mathcal{F}$ are <i>disjoint</i>, $\mathbf{P}(\bigcup_i A_i) = \sum_i \mathbf{P}(A_i)$.
Conditioning, Independence & Prob. Ineq.s
Conditioning Definition: For $\mathbf{P}(B) \neq 0, \mathbf{P}(A B) = \mathbf{P}(A \cap B)/\mathbf{P}(B)$. - Nonnegativity: $\mathbf{P}(A B) \geq 0, \forall A \in \mathcal{F}$, - Normalization: $\mathbf{P}(\Omega B) = 1$, - Countable additivity: If $A_i \in \mathcal{F}$ are <i>disjoint</i>, $\mathbf{P}(\bigcup_i A_i B) = \sum_i \mathbf{P}(A_i B)$. Multiplication Rule: $\mathbf{P}(A \cap B \cap C) = \mathbf{P}(A) \cdot \mathbf{P}(B A) \cdot \mathbf{P}(C A \cap B)$. Total Probability Theorem: For a partition of sample space $A_1, A_2, \dots, A_n$, assuming $\mathbf{P}(A_i) > 0$ for all i, $\mathbf{P}(B) = \sum_{i=1}^n \mathbf{P}(A_i) \mathbf{P}(B A_i).$ Bayes’s Rule: $\mathbf{P}(A_i B) = \frac{\mathbf{P}(A_i) \mathbf{P}(B A_i)}{\sum_{j=1}^n \mathbf{P}(A_j) \mathbf{P}(B A_j)}$. Independence Definition: Event A and B are called independent if $\mathbf{P}(A \cap B) = \mathbf{P}(A) \mathbf{P}(B)$. - For $\mathbf{P}(B) > 0, \mathbf{P}(A B) = \mathbf{P}(A)$. Conditional Independence: $\mathbf{P}(A \cap B C) = \mathbf{P}(A C) \mathbf{P}(B C)$. Collective Independence: Events $A_1, A_2, \dots, A_n$ are called independent if $\mathbf{P}(A_i \cap A_j \cap \dots \cap A_q) = \mathbf{P}(A_i) \mathbf{P}(A_j) \dots \mathbf{P}(A_q),$ for any distinct indices $i, j, \dots, q$ chosen from $\{1, 2, \dots, n\}$. Inclusion-Exclusion Principle: $\mathbf{P}(\bigcup_{i=1}^n A_i) = \sum \mathbf{P}(A_i) - \sum_{i < j} \mathbf{P}(A_i \cap A_j) + \dots + (-1)^{n-1} \mathbf{P}(A_1 \cap \dots \cap A_n)$. Hat Problem (Derangement): - Derangement Number: $D_n = n! \sum_{i=0}^n (-1)^i / i!$. $D_n = n D_{n-1} + (-1)^n$. $\lim_{n \rightarrow \infty} D_n / n! = 1/e$. * Prob. of at least one match is $\approx 1 - 1/e$ as $n \rightarrow \infty$. Symmetric Difference: $\mathbf{P}(A \triangle B) = \mathbf{P}(A) + \mathbf{P}(B) - 2\mathbf{P}(A \cap B)$. Probability Inequalities Fréchet Inequalities: - Intersection: $\max(0, \mathbf{P}(A) + \mathbf{P}(B) - 1) \leq \mathbf{P}(A \cap B) \leq \min(\mathbf{P}(A), \mathbf{P}(B))$.

- Union: $\max(\mathbf{P}(A), \mathbf{P}(B)) \leq \mathbf{P}(A \cup B) \leq \min(1, \mathbf{P}(A) + \mathbf{P}(B))$.
- **Boole’s and Bonferroni Inequalities:** $\mathbf{P}(\bigcup_{i=1}^n A_i) \leq \sum_{i=1}^n \mathbf{P}(A_i)$, $\mathbf{P}(\bigcap_{i=1}^n A_i) \geq \sum_{i=1}^n \mathbf{P}(A_i) - (n - 1)$
- **Union of independent events:** $\mathbf{P}(\bigcup_{i=1}^n A_i) = 1 - \prod_{i=1}^n (1 - \mathbf{P}(A_i)) \geq 1 - \exp(-\sum_{i=1}^n \mathbf{P}(A_i))$.

Discrete Random Variable
● Random Variable: Given a probability space $(\Omega, \mathcal{F}, \mathbf{P})$, a random variable is a <i>function</i> $X: \Omega \rightarrow \mathbb{R}$ such that $X^{-1}(A) \in \mathcal{F}$ for each $A \in \mathcal{B}(\mathbb{R})$. ● PMF: $p_X(x) = \mathbf{P}(X = x)$. ● Derived PMF: $Y = g(X) \Rightarrow p_Y(y) = \sum_{x: g(x)=y} p_X(x)$. ● Expectation: $\mathbf{E}[X] = \sum_i x_i p_X(x_i)$. ● Variance: $\text{var}[X] = \mathbf{E}\left[(X - \mathbf{E}[X])^2\right] = \mathbf{E}[X^2] - (\mathbf{E}[X])^2$. ● $\mathbf{E}[aX + \beta] = a\mathbf{E}[X] + \beta, \mathbf{E}[X + Y] = \mathbf{E}[X] + \mathbf{E}[Y]$. ● $\text{var}[aX + \beta] = a^2 \text{var}[X]$. ● Geometric RV: * $p_X(k) = (1 - p)^{k-1} p$, * $\mathbf{E}[X] = 1/p$, * $\mathbf{E}[X^2] = (2 - p)/p^2$, * $\text{var}[X] = (1 - p)/p^2$. * Memorylessness: $p_X(k X > n) = (1 - p)^{k-n-1} p; \mathbf{E}[X X > n] = 1/p + n$. □ Bernoulli RV: * $p_X(k) = \begin{cases} p, & k = 1, \\ 1 - p, & k = 0, \end{cases}$ * $\mathbf{E}[X] = p$, * $\mathbf{E}[X^2] = p$, * $\text{var}[X] = p(1 - p)$. □ Discrete Uniform RV: * $p_X(k) = 1/(b - a + 1)$, if $k = a, a + 1, \dots, b$, * $\mathbf{E}[X] = (a + b)/2$, * $\text{var}[X] = (b - a)(b - a + 2)/12$. □ Binomial RV: * $p_X(k) = C_n^k p^k (1 - p)^{n-k}$, * $\mathbf{E}[X] = np$, * $\mathbf{E}[X^2] = np(1 - p + np)$, * $\text{var}[X] = np(1 - p)$. □ Poisson RV: $\lambda = np$ as $n \rightarrow \infty$ in binomial RV. * $p_X(k) = e^{-\lambda} \lambda^k / k!, k = 0, 1, \dots$, * $\mathbf{E}[X] = \lambda$, * $\mathbf{E}[X^2] = \lambda + \lambda^2$, * $\text{var}[X] = \lambda$, * Independent sum: $X_1 + X_2 \sim \text{Poisson}(\lambda_1 + \lambda_2)$.
Multiple Discrete RV
● Conditioning: □ Conditional PMF: $p_{X A}(x) = \mathbf{P}\{X = x\} \cap A) / \mathbf{P}(A), p_{X Y}(x y) = p_{X,Y}(x, y) / p_Y(y)$. □ Conditional Expectation: $\mathbf{E}[X A] = \sum_x x p_{X A}(x)$. □ Total Expectation Theorem: $\mathbf{E}[X] = \mathbf{P}(A_1) \mathbf{E}[X A_1] + \dots + \mathbf{P}(A_n) \mathbf{E}[X A_n]$. □ Marginal PMF: $p_{X,Y}(x, y) = \sum_z p_{X,Y,Z}(x, y, z); p_X(x) = \sum_y p_{X,Y}(x, y)$. ● Independence: * $\forall x, y: p_{X,Y}(x, y) = p_X(x) p_Y(y)$. ● For Independent RVs: □ $f(X)$ and $g(Y)$ are also independent. □ $\mathbf{E}[XY] = \mathbf{E}[X] \mathbf{E}[Y]$. □ $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y), \text{var}(X - Y) = \text{var}(X) + \text{var}(Y)$. □ Convolution: $W = X + Y \Rightarrow p_W(w) = \sum_x p_X(x) p_Y(w - x)$. ● Conditional Independence: * $\forall x, y: p_{X,Y}(x, y A) = p_X(x A) p_Y(y A)$.

Continuous RV
● PDF: $\mathbf{P}(a \leq X \leq b) = \int_a^b f_X(x) dx$; $\mathbf{P}(x \leq X \leq x + \delta) \approx f_X(x) \delta$. ● Expectation: $\mathbf{E}[X] = \int_{-\infty}^{+\infty} x f_X(x) dx$. ● Variance: $\text{var}(X) = \int_{-\infty}^{+\infty} (x - \mathbf{E}[X])^2 f_X(x) dx = \mathbf{E}[X^2] - (\mathbf{E}[X])^2$. ● Uniform RV: * $f_X(x) = 1/(b - a), a \leq x \leq b$, * $\mathbf{E}[X] = (a + b)/2$, * $\mathbf{E}[X^2] = (a^2 + ab + b^2)/3$, * $\text{var}[X] = (b - a)^2 / 12$. □ Exponential RV: * $f_X(x) = \lambda e^{-\lambda x}, x \geq 0$, * $F_X(x) = 1 - e^{-\lambda x}, x \geq 0$. * $\mathbf{E}[X] = 1/\lambda$, * $\mathbf{E}[X^2] = 2/\lambda^2$, * $\text{var}[X] = 1/\lambda^2$. * $\mathbf{P}(X_1 < X_2) = \lambda_1 / (\lambda_1 + \lambda_2)$. * Memorylessness: $f_X(x X > a) = \lambda e^{-\lambda(x-a)}, \mathbf{P}(X \geq c + x X \geq c) = \mathbf{P}(X \geq x) = e^{-\lambda x}$; □ Gaussian RV ($X \sim N(\mu, \sigma^2)$): * $f_X(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-(x-\mu)^2/(2\sigma^2)},$ * $F_X(\lambda) = \Phi(\lambda), \Lambda \sim N(0, 1)$. * $\mathbf{E}[X] = \mu$, * $\mathbf{E}[X^2] = \mu^2 + \sigma^2$, * $\mathbf{E}[X^4] = 3\sigma^4 + 6\sigma^2 \mu^2 + \mu^4$, * $\text{var}[X] = \sigma^2$. * $Y = aX + \beta \sim N(a\mu + \beta, a^2 \sigma^2)$. ● CDF: □ $F_X(x) = \int_{-\infty}^{x+} f_X(t) dt$. □ $F_X(x)$ discontinuous $\Rightarrow X$ has discrete part. □ $\mathbf{P}(a < X \leq b) = F_X(b) - F_X(a)$. □ $\mathbf{E}[X] = \int_0^{+\infty} (1 - F_X(x)) dx - \int_0^{-\infty} F_X(x) dx$. ● Derived PDF: □ CDF Method: find $F_Y(y) = \mathbf{P}(g(X) \leq y) \Rightarrow f_Y(y) = F_Y'(y)$. □ Linear: $Y = aX + b \Rightarrow f_Y(y) = \frac{1}{ a } f_X\left(\frac{y-b}{a}\right)$. □ Monotone: $y = g(x) \Leftrightarrow x = h(y) \Rightarrow f_Y(y) = f_X(h(y)) h'(y)$. ● Min and Max of i.i.d. RVs ($X_1, \dots, X_n$): □ Max: $Z = \max(X_i) \Rightarrow F_Z(z) = [F_X(z)]^n \Rightarrow f_Z(z) = n[f_X(z)]^{n-1} f_X(z)$. □ Min: $W = \min(X_i) \Rightarrow F_W(w) = 1 - [1 - F_X(w)]^n$. □ Shortcut: If $X_i \sim \text{Exp}(\lambda_i)$, then $\min(X_1, \dots, X_n) \sim \text{Exp}(\sum \lambda_i)$. ● 2D Uniform & Geometric Probability: □ $f_{X,Y}(x, y) = 1/\text{Area}(S)$ for $(x, y) \in S$. □ Broken Stick / Triangle Inequality: Break a line into 3 pieces. They form a triangle iff. all pieces are shorter than 1/2. Prob. is 1/4.
Multiple Continuous RV
Conditioning: ● Conditional PDF: $f_{X Y}(x y) = f_{X,Y}(x, y) / f_Y(y)$. ● Conditional Expectation: $\mathbf{E}[X Y = y] = \int_{-\infty}^{+\infty} x f_{X Y}(x y) dx$. ● Total Expectation Theorem: $\mathbf{E}[X] = \int_{-\infty}^{+\infty} \mathbf{E}[X Y = y] f_Y(y) dy$. ● Marginal PDF: $f_X(x) = \int_{-\infty}^{+\infty} f_{X,Y}(x, y) dy$. Independence: ● $\forall x, y: f_{X,Y}(x, y) = f_X(x) f_Y(y)$. ● For independent RVs: □ $f(X)$ and $g(Y)$ are also independent, □ $\mathbf{E}[XY] = \mathbf{E}[X] \mathbf{E}[Y]$, □ $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$. □ Convolution: If X and Y are independent and $Z = X + Y, f_Z(z) = (f_X * f_Y)(z) = \int_{-\infty}^{+\infty} f_X(x) f_Y(z - x) dx$. □ $X \sim N(\mu_X, \sigma_X^2), Y \sim N(\mu_Y, \sigma_Y^2) \Rightarrow X + Y \sim N(\mu_X + \mu_Y, \sigma_X^2 + \sigma_Y^2)$.

- **Joint / Multiple CDF:** $F_{X,Y}(x, y) = \mathbf{P}(X \leq x, Y \leq y)$.
- **Marginals:** $F_X(x) = F_{X,Y}(x, \infty), F_Y(y) = F_{X,Y}(\infty, y)$.
- **From/To PDF:** $F_{X,Y}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(u, v) dv du \Rightarrow f_{X,Y}(x, y) = \frac{\partial^2}{\partial x \partial y} F_{X,Y}(x, y)$.
- **Independence:** $X \perp Y \iff F_{X,Y}(x, y) = F_X(x) F_Y(y)$.
- **Conditional CDF:**
- **Given Event** $A: F_{X|A}(x) = \mathbf{P}(X \leq x | A) = \frac{\mathbf{P}(\{X \leq x\} \cap A)}{\mathbf{P}(A)}$.
- **Given** $Y = y: F_{X|Y}(x|y) = \int_{-\infty}^x f_{X|Y}(t|y) dt$.
- **Derivative:** $f_{X|Y}(x|y) = \frac{\partial}{\partial x} F_{X|Y}(x|y)$.
- **Total Prob:** $F_X(x) = \sum \mathbf{P}(A_i) F_{X|A_i}(x)$.
- **Continuous Bayes’ Rule:**

$$f_{X|Y}(x|y) = \frac{f_{Y|X}(y|x) f_X(x)}{f_Y(y)} = \frac{f_{Y|X}(y|x) f_X(x)}{\int_{-\infty}^{\infty} f_{Y|X}(y|t) f_X(t) dt},$$

$$\mathbf{P}(A|Y = y) = \frac{\mathbf{P}(A) f_{Y|A}(y)}{f_Y(y)}. \text{(Mixed Bayes)}$$

Covariance and Correlation
● Covariance: $\text{cov}(X, Y) = \mathbf{E}[(X - \mathbf{E}[X])(Y - \mathbf{E}[Y])] = \mathbf{E}[XY] - \mathbf{E}[X] \mathbf{E}[Y]$. ● Uncorrelated: $\text{cov}(X, Y) = 0$; independent $\Rightarrow \text{cov}(X, Y) = 0$. ● $\text{cov}(X, X) = \text{var}(X); \text{cov}(X, aY + b) = a \text{cov}(X, Y); \text{cov}(X, Y + Z) = \text{cov}(X, Y) + \text{cov}(X, Z)$. ● $\text{var}(X + Y) = \text{var}(X) + \text{var}(Y) + 2 \text{cov}(X, Y)$. ● $\text{var}(\sum_i X_i) = \sum_i \text{var}(X_i) + \sum_{i \neq j} \text{cov}(X_i, X_j)$. ● Correlation: $\rho(X, Y) = \frac{\text{cov}(X, Y)}{\sqrt{\text{var}(X) \text{var}(Y)}}, -1 \leq \rho \leq 1$. ● $\rho = 1 \iff X - \mathbf{E}[X] = c(Y - \mathbf{E}[Y])$.
Conditional Expectation
● Iterated Expectation: $\mathbf{E}[X] = \mathbf{E}[\mathbf{E}[X Y]]$. □ Y discrete: $\mathbf{E}[\mathbf{E}[X Y]] = \sum_g \mathbf{E}[X Y = g] p_Y(g)$. □ Y continuous: $\mathbf{E}[\mathbf{E}[X Y]] = \int \mathbf{E}[X Y = y] f_Y(y) dy$. ● Estimator: $\hat{X} = \mathbf{E}[X Y], \tilde{X} = X - \hat{X}$. ● $\mathbf{E}[\tilde{X} Y] = 0 \Rightarrow \mathbf{E}[\tilde{X}] = 0; \text{cov}(\hat{X}, \tilde{X}) = 0$. ● Conditional Variance: $\text{var}(X Y) = \mathbf{E}[(X - \mathbf{E}[X Y])^2 Y]$. ● Total Variance: $\text{var}(X) = \text{var}(\mathbf{E}[X Y]) + \mathbf{E}[\text{var}(X Y)]$.
Transforms
● Definition: $\mathcal{M}_X(s) = \mathbf{E}[e^{sX}]$; - discrete $\sum_x p_X(x) e^{sx}$; continuous $\int f_X(x) e^{sx} dx$. ● Recover Law: MGF exists near $0 \Rightarrow$ unique distribution. □ Discrete: expand $\mathcal{M}_X(s) = \sum_x p_X(x) e^{sx}$; coeff. of e^{sx} is $p_X(x)$; - or $p_X(x) = \mathcal{Z}^{-1}\{\mathcal{M}_X(-\ln z)\}$. □ Continuous: $f_X(x) = \mathcal{L}_b^{-1}\{\mathcal{M}_X(-s)\}$; - or $f_X(x) = \mathcal{F}^{-1}\{\mathcal{M}_X(it)\} = \frac{1}{2\pi} \int e^{-itx} \mathcal{M}_X(it) dt$. ● $\mathcal{M}_X(0) = 1; \mathbf{E}[X^n] = \left. \frac{d^n \mathcal{M}_X(s)}{ds^n} \right _{s=0}$. ● $Y = aX + b \Rightarrow \mathcal{M}_Y(s) = e^{bs} \mathcal{M}_X(as)$. ● Independent $Z = \sum_{i=1}^n X_i \Rightarrow \mathcal{M}_Z(s) = \prod_i \mathcal{M}_{X_i}(s)$. ● Mixture: $f_Y = \sum_i p_i f_{X_i} \Rightarrow \mathcal{M}_Y(s) = \sum_i p_i \mathcal{M}_{X_i}(s)$. ● Random Sum: $Y = X_1 + \dots + X_N, \text{i.i.d. } X_i \perp N$: □ $\mathbf{E}[Y] = \mathbf{E}[N] \mathbf{E}[X]$, □ $\text{var}(Y) = \mathbf{E}[N] \text{var}(X) + (\mathbf{E}[X])^2 \text{var}(N)$, □ $\mathcal{M}_Y(s) = \mathcal{M}_N(\ln \mathcal{M}_X(s))$. ● Common $\mathcal{M}_X(s)$: □ Bernoulli: $1 - p + pe^s$; Binomial: $(1 - p + pe^s)^n$, □ Geometric: $\frac{pe^s}{1 - (1 - p)e^s}$; Poisson: $\exp(\lambda(e^s - 1))$, □ Discrete uniform $\{a, \dots, b\}: \frac{e^{as}(1 - e^{(b-a+1)s})}{(b - a + 1)(1 - e^s)}$, □ Uniform $[a, b]: \frac{e^{bs} - e^{as}}{(b - a)s}$; Exponential: $\frac{\lambda}{\lambda - s}$, □ Gaussian $N(\mu, \sigma^2): \exp(i\mu s + \sigma^2 s^2 / 2)$.

□ Pascal: $\left(\frac{pe^s}{1-(1-p)e^s}\right)^k$; NegBin: $\left(\frac{p}{1-(1-p)e^s}\right)^k$.
RV Inequalities
● Markov: $X \geq 0, a > 0 \Rightarrow \mathbf{P}(X \geq a) \leq \mathbf{E}[X]/a$. □ “=” iff $X \in \{0, a\}$. ● Chebyshev: $\mathbf{P}(X - \mathbf{E}[X] \geq c) \leq \text{var}(X)/c^2$. □ “=” iff $X \in \{\mu - c, \mu, \mu + c\}$, $\mathbf{P}(X = \mu \pm c) = \sigma^2/(2c^2)$. ● Variance Bound: $X \in [a, b] \Rightarrow \text{var}(X) \leq (b - a)^2/4$. □ “=” iff $\mathbf{P}(X = a) = \mathbf{P}(X = b) = 1/2$. ● Chernoff: □ Right-tail: $\mathbf{P}(X \geq a) \leq \inf_{s \geq 0} \text{e}^{-sa} \mathcal{M}_X(s) = \text{e}^{-\sup_{s \geq 0} (sa - \ln \mathcal{M}_X(s))}$, □ Left-tail: $\mathbf{P}(X \leq a) \leq \inf_{s \leq 0} \text{e}^{-sa} \mathcal{M}_X(s) = \text{e}^{-\sup_{s \leq 0} (sa - \ln \mathcal{M}_X(s))}$.
LLN and CLT
● WLLN: $M_n = \frac{1}{n} \sum_{i=1}^n X_i$, i.i.d., $\mathbf{E}[X_i] = \mu$, $\text{var}(X_i) = \sigma^2$. □ $\mathbf{E}[M_n] = \mu$, $\text{var}(M_n) = \sigma^2/n$, □ $\mathbf{P}(M_n - \mu \geq \epsilon) \leq \sigma^2/(n\epsilon^2) \rightarrow 0$, □ $M_n \xrightarrow{\text{P}} \mu$. ● Convergence: □ In Probability: $X_n \xrightarrow{\text{P}} X \Leftrightarrow \mathbf{P}(X_n - X \geq \epsilon) \rightarrow 0, \forall \epsilon > 0$, □ In Distribution: $X_n \xrightarrow{D} X \Leftrightarrow F_{X_n}(x) \rightarrow F_X(x)$ at continuous x, □ Almost Surely: $X_n \xrightarrow{\text{a.s.}} X \Rightarrow \mathbf{P}(\lim_n X_n = X) = 1$, □ “Almost Surely”: For almost all sample paths ω. □ r-th Mean: $X_n \xrightarrow{L} X \Leftrightarrow \mathbf{E}[X_n - X ^r] \rightarrow 0$, □ Relations: a.s. $\Rightarrow \mathbf{P} \Rightarrow D$; r-mean $\Rightarrow \mathbf{P} \Rightarrow D$; $r \geq s \Rightarrow L^r \Rightarrow L^s$. ● CLT: $S_n = \sum_{i=1}^n X_i$, i.i.d., $\mathbf{E}[X_i] = \mu$, $\text{var}(X_i) = \sigma^2$. □ $Z_n = \frac{S_n - n\mu}{\sigma\sqrt{n}} \xrightarrow{D} N(0, 1)$; $\mathbf{P}(Z_n \leq z) \rightarrow \Phi(z)$.
● Transform: $\mathcal{M}_{Z_n}(s) = \text{e}^{-s\mu\sqrt{n}/\sigma} \left(\mathcal{M}_{X_1}\left(\frac{s}{\sigma\sqrt{n}}\right)\right)^n \rightarrow \text{e}^{s^2/2}$.
● Normal Approx.: $S_n \approx N(n\mu, n\sigma^2)$; $\mathbf{P}(S_n \leq c) \approx \Phi\left(\frac{c - n\mu}{\sigma\sqrt{n}}\right)$.
● Binomial Approx.: $X_i \sim \text{Bernoulli}(p)$, $S_n = \sum_i X_i \sim \text{Bin}(n, p)$, $q = 1 - p$. □ $\frac{S_n - np}{\sqrt{npq}} \approx N(0, 1)$, □ $\mathbf{P}(k \leq S_n \leq l) \approx \Phi\left(\frac{l + 0.5 - np}{\sqrt{npq}}\right) - \Phi\left(\frac{k - 0.5 - np}{\sqrt{npq}}\right)$.
● SLLN: $M_n = S_n/n \Rightarrow \mathbf{P}(\lim_{n \rightarrow \infty} M_n = \mu) = 1$, i.e. $M_n \xrightarrow{\text{a.s.}} \mu$.
Event Limits and Borel–Cantelli
● Set Limits: □ $\limsup A_n = \bigcap_{k \geq 1} \bigcup_{n \geq k} A_n = \{A_n \text{ i.o.}\}$; □ “infinite often”= $\forall N \geq 0, \exists n \geq N, A_n$ happens. □ $\liminf A_n = \bigcup_{k \geq 1} \bigcap_{n \geq k} A_n = \{A_n \text{ eventually}\}$. □ “eventually”= $\exists N \geq 0, \forall n \geq N, A_n$ happens. □ $\liminf A_n \subseteq \limsup A_n$; “=” $\Rightarrow \lim A_n$ exists. ● Monotone Convergence Theorem: $0 \leq X_n \uparrow X \Rightarrow \mathbf{E}[X_n] \uparrow \mathbf{E}[X]$; events $A_n \uparrow A \Rightarrow \mathbf{P}(A_n) \uparrow \mathbf{P}(A)$. □ Let X_n = the sum of an RV seq. $\{Y_i\}$, $Y_i \geq 0$, $X = \lim_n X_n$, $\mathbf{E}[X] < \infty \Rightarrow \mathbf{P}(X = \infty) = 0 \Rightarrow X$ converge a.s.. □ $X = \sum_{i=1}^\infty Y_i$ converges a.s. $\Rightarrow \{Y_i\} \xrightarrow{\text{a.s.}} 0$. ● Borel–Cantelli I: $\sum_n \mathbf{P}(A_n) < \infty \Rightarrow \mathbf{P}(A_n \text{ i.o.}) = 0$; ● Borel–Cantelli II: independent A_n, $\sum_n \mathbf{P}(A_n) = \infty \Rightarrow \mathbf{P}(A_n \text{ i.o.}) = 1$. ● a.s. Criterion: $X_n \xrightarrow{\text{a.s.}} X \Leftrightarrow \forall \epsilon > 0, \mathbf{P}(X_n - X > \epsilon \text{ i.o.}) = 0$. □ Equiv.: $\lim_{n \rightarrow \infty} \mathbf{P}(\sup_{k \geq n} X_k - X > \epsilon) = 0, \forall \epsilon > 0$, □ Enough: $\forall m \geq 1, \sum_n \mathbf{P}(X_n - X > 1/m) < \infty$. ● Subsequence: $X_n \xrightarrow{\text{P}} X \Rightarrow \exists n_k : X_{n_k} \xrightarrow{\text{a.s.}} X$; □ proof: choose $\mathbf{P}(X_{n_k} - X > 1/k) < 2^{-k}$. ● Kolmogorov Ineq.: independent RVs X_i, $\mathbf{E}[X_i] = 0$, $\text{var}(X_i) < \infty$, $S_k = \sum_{i=1}^k X_i \Rightarrow \mathbf{P}(\max_{k \leq n} S_k \geq a) \leq \text{var}(S_n)/a^2 = \sum_{i=1}^n \text{var}(X_i)/a^2$. ● Cantelli SLLN: i.i.d., $\mathbf{E}[X^4] < \infty \Rightarrow \mathbf{E}[(\sum_{i=1}^n (X_i - \mu))^4] = O(n^2)$; Borel–Cantelli I $\Rightarrow S_n/n \xrightarrow{\text{a.s.}} \mu$. ● $\limsup A_n$ is event; $\limsup \mathbf{P}(A_n)$ is number;

$\mathbf{P}(\limsup A_n) \geq \limsup \mathbf{P}(A_n)$.

Stochastic Processes

- Stochastic Process**: A collection of RVs indexed by time.
 - View as a mapping: $Z: T \times \Omega \rightarrow \mathbb{R}; (t, \omega) \mapsto Z_t(\omega)$.
 - View as a sequence: $\{Z_{t_1}, Z_{t_2}, \dots\}$, $t_1, t_2 \in T$ is a sequence of time-relevant RVs defined on prob. space $(\Omega, \mathcal{F}, \mathbf{P})$.
 - Once the sample path ω is fixed, values of the whole RV walk $\{Z_{t_1}(\omega), Z_{t_2}(\omega), \dots\}$ are fixed.
- Bernoulli Process**: discrete-time discrete-value SP $\{X_i\}_{i \geq 1}$; $X_i \sim$ i.i.d. Bernoulli(p).
 - Intuition: keep tossing a coin and record if i -th round is a head as X_i .
 - Independence: seq. $X_{n+1}, X_{n+2}, \dots$ after n is a Bernoulli SP independent of $X_1, \dots, X_n$.
 - Memoryless: W_n = trials from n to next success $\sim \text{Geo}(p)$,
 - Interarrival: Y_k = time of k -th success, $T_1 = Y_1, T_k = Y_k - Y_{k-1}$; $T_i \sim$ i.i.d. $\text{Geo}(p)$; $Y_k = \sum_{i=1}^k T_i$,
 - Trials until k -th success: $Y_k \sim \text{Pascal}(k, p)$,
 $*p_{Y_k}(t) = \text{C}_{t-1}^{k-1} p^k (1-p)^{t-k}, t = k, k+1, \dots$,
 $*\mathbf{E}[Y_k] = k/p, \text{var}(Y_k) = k(1-p)/p^2$,
 - Failures until k -th success: $Z_k \sim \text{NegBin}(k, p)$,
 $*p_{Z_k}(t) = \text{C}_{t+k-1}^{k-1} p^k (1-p)^t, t = 0, 1, 2, \dots$,
 $*\mathbf{E}[Z_k] = k(1-p)/p, \text{var}(Z_k) = k(1-p)/p^2$,
 - Splitting: keep w.p. $q \Rightarrow$ Bernoulli SP(pq),
 - Merging Independent Bernoullis: Bernoulli(p) OR Bernoulli(q) $\Rightarrow$ Bernoulli($p+q-pq$).
- Poisson Process**: continuous-time discrete-value SP $\{N_t\}_{t \geq 0}$, $N_0 = 0$, $N_t \sim \text{Poisson}(\lambda t)$; .
 - Intuition: count the number of arrivals w.p. $\lambda \delta$ for a small interval δ from 0 to t as N_t .
 - Homogeneity: $\mathbf{P}(k, t) = \mathbf{P}(k \text{ arrivals in interval of duration } t) = \sum_{n=0}^{\infty} \mathbf{P}(N_{t+T} = n+k | N_T = n) \mathbf{P}(n, T)$
 - Independent: Numbers of arrivals in disjoint time intervals are independent.
 - Small δ : $P(0, \delta) = 1 - \lambda \delta + o(\delta), P(1, \delta) = \lambda \delta + o(\delta), P(k, \delta) = o(\delta), k > 1$,
 - Bernoulli Relation: Select time infinitesimal δ , $n = t/\delta, p = \lambda \delta$; $\{X_i\}$ is a Bernoulli SP(p), $S_n = \sum_{k=1}^n X_i \sim \text{Binomial RV}(n, p) \rightarrow N_t \sim \text{Poisson RV}(\lambda t), n \rightarrow \infty$,
 - First Arrival: $Y_1 = T_1 \sim \text{Exp}(\lambda)$; $F_{Y_1}(t) = 1 - e^{-\lambda t}, f_{Y_1}(t) = \lambda e^{-\lambda t}$;
 - Interarrival: Y_k = time of k th arrival, $T_1 = Y_1, T_k = Y_k - Y_{k-1}$; $T_i \sim$ i.i.d. $\text{Exp}(\lambda)$, $Y_k = \sum_{i=1}^k T_i$,
 - k -th arrival: $Y_k \sim \text{Erlang}(\lambda, k)$,
 $*f_{Y_k}(y) = \lambda^k y^{k-1} e^{-\lambda y} / (k-1)!, y \geq 0$;
 $*\mathbf{E}[Y_k] = k/\lambda, \text{var}(Y_k) = k/\lambda^2$,
 - Splitting: keep w.p. $p \Rightarrow \text{Poisson}(\lambda p)$,
 - Merging Independent Poissons: $\text{Poisson}(\lambda_1) + \text{Poisson}(\lambda_2) = \text{Poisson}(\lambda_1 + \lambda_2)$;
 next arrival from SP 1 w.p. $\lambda_1/(\lambda_1 + \lambda_2)$.
- Random Sum**: $Y = \sum_{i=1}^N X_i, Y = 0$ if $N = 0$; $\mathcal{M}_Y(s) = \mathcal{M}_N(\ln \mathcal{M}_X(s))$.
- Random Sum Table**:

X_i	N	Y
Bernoulli(p)	Binomial(m, q)	Binomial(m, pq)
Bernoulli(p)	Poisson(λ)	Poisson(λp)
Bernoulli(p)	NegBin(r, q)	NegBin($r, q/(q+p-pq)$)
Geo(p)	Geo(q)	Geo(pq)
Exponential(λ)	Geo(q)	Exponential(λq)

Entropy
● Discrete: $H(X) = \mathbf{E}[-\ln p_X(X)] = -\sum_k p_X(x_k) \ln p_X(x_k)$. ● Continuous: $h(X) = \mathbf{E}[-\ln f_X(X)] = -\int f_X(x) \ln f_X(x) \text{d}x$. ● Bernoulli: $H(X) = -p \log_2 p - (1 - p) \log_2(1 - p)$; max 1 at $p = 1/2$, min 0 at $p = 0, 1$. ● Max Entropy: □ finite $\{x_1, \dots, x_n\} \Rightarrow p_k = 1/n$, □ $[a, b] \Rightarrow U(a, b)$, □ $(0, \infty)$, $\mathbf{E}[X] = 1/\lambda \Rightarrow \text{Exp}(\lambda)$, □ $\mathbb{R}$, fixed $\mu, \sigma^2 \Rightarrow N(\mu, \sigma^2)$.

Integration Formula
● $\int x \text{e}^{ax} \text{d}x = \frac{1}{a^2}(ax - 1)\text{e}^{ax}, \int x^2 \text{e}^{ax} \text{d}x = \frac{a^2 x^2 - 2ax + 2}{a^3} \text{e}^{ax}, \int x^3 \text{e}^{ax} \text{d}x = \frac{a^3 x^3 - 3a^2 x^2 + 6ax - 6}{a^4} \text{e}^{ax}, \int x^n \text{e}^{ax} \text{d}x = \frac{1}{a} x^n \text{e}^{ax} - \frac{n}{a} \int x^{n-1} \text{e}^{ax} \text{d}x$. ● $\int_0^\infty x^n \text{e}^{-ax} \text{d}x = n!/a^{n+1}$. ● $\int_a^\infty x^n \text{e}^{-\lambda x} \text{d}x = \text{e}^{-\lambda a} \sum_{j=0}^n \frac{n! a^j}{j! \lambda^{n+1-j}}$. ● $\int_a^\infty \lambda \text{e}^{-\lambda x} \text{d}x = \text{e}^{-\lambda a}$; $\int_0^a \lambda \text{e}^{-\lambda x} \text{d}x = 1 - \text{e}^{-\lambda a}$. ● $\int_a^\infty \frac{\lambda^k x^{k-1} \text{e}^{-\lambda x}}{(k-1)!} \text{d}x = \text{e}^{-\lambda a} \sum_{j=0}^{k-1} \frac{(\lambda a)^j}{j!}$. ● $\int \sqrt{a^2 - x^2} \text{d}x = \frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \arcsin \frac{x}{a}$. ● $\int \ln x \text{d}x = x \ln x - x, \int x^n \ln x \text{d}x = \frac{1}{n+1} x^{n+1} \left(\ln x - \frac{1}{n+1}\right)$. ● $\int x \exp\left(-x^2/\sigma^2\right) \text{d}x = -\frac{\sigma^2}{2} \exp\left(-x^2/\sigma^2\right) + C$. ● $\int_{-\infty}^{+\infty} \exp\left(-(x - \lambda)^2/\mu^2\right) \text{d}x = \mu \sqrt{\pi}$. ● $\int_{-\infty}^{+\infty} x^2 \exp\left(-x^2/\mu^2\right) \text{d}x = \frac{\mu^3}{2} \sqrt{\pi}$.